

Summary: *The Purchasing Power Parity Debate*

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1	Paper Overview	2
1.1	Introduction & Context	2
1.2	Structure of the Paper	2
1.3	Conclusions	2
1.4	Key Takeaways	3
2	Key Points	4
2.1	The PPP Concept: Absolute vs. Relative	4
2.2	Informal Empirical Evidence: Figures 1 and 2	4
2.3	The Monetary Approach and the Collapse of Continuous PPP	4
2.4	Random Walks, Unit Roots, and the Failure to Find Long-Run PPP	5
2.5	The Power Problem	7
2.6	Long-Span Studies	7
2.7	Panel-Data Studies and Their Limitations	7
2.8	The Two PPP Puzzles	8
2.9	Nonlinear Adjustment and Transactions Costs	8
2.10	Short-Run Exchange Rate Volatility	9
2.11	The Harrod-Balassa-Samuelson Effect and Trending Equilibrium	9
2.12	Net Foreign Asset Positions and the Real Exchange Rate	9
2.13	Engel's Decomposition and Unifying the Puzzles	10
3	Logical Chains	11
3.1	Chain 1: From Transactions Costs to Resolution of the PPP Puzzles	11
3.2	Chain 2: The Harrod-Balassa-Samuelson Mechanism	11
4	Key Terms & Definitions	13

1 Paper Overview

1.1 Introduction & Context

Purchasing power parity (PPP) is one of the oldest and most intuitive theories of exchange rate determination. It states that the nominal exchange rate between two currencies should equal the ratio of their aggregate price levels, so that a unit of currency buys the same basket of goods at home and abroad. Despite its conceptual simplicity, PPP has been the subject of intense empirical controversy for over three decades.

The paper surveys the PPP debate from the early 1970s—when the collapse of the Bretton Woods system ushered in floating exchange rates—through the early 2000s. The central tension is between (a) the theoretical attractiveness of PPP as a long-run anchor for real exchange rates and (b) the persistent difficulty of finding strong empirical support for it, especially in the post-Bretton Woods floating-rate era. Two specific “PPP puzzles” frame the discussion: the puzzling *weakness* of evidence for long-run PPP, and the puzzling *slowness* (half-lives of 3–5 years) of real exchange rate reversion toward PPP even when mean reversion is detected.

1.2 Structure of the Paper

1. **Introduction and informal evidence (Sections 1–2):** The concept of PPP—absolute versus relative—is introduced, along with graphical evidence from long time series and cross-country panels showing that PPP holds better over longer horizons and for producer prices.
2. **Three decades of the PPP debate (Section 3):** A chronological tour covering the monetary approach and continuous PPP (1970s), the failure of unit root tests (1980s), and the power problem.
3. **Responses to the power problem (Sections 4–5):** Long-span studies, panel-data studies, and their respective limitations.
4. **PPP puzzles and nonlinearity (Section 6):** Identification of the two PPP puzzles and the nonlinear (threshold/STAR) approach to resolving them.
5. **Remaining puzzles—disturbances and long-run equilibrium (Sections 7–8):** Short-run exchange rate volatility, the role of nontraded goods, the Harrod-Balassa-Samuelson effect, and wealth/net-asset effects on the equilibrium real exchange rate.
6. **Conclusion (Section 9):** Synthesis showing how theory and empirics are converging, with suggestions for future research.

1.3 Conclusions

The paper concludes that the gap between PPP theory and empirical evidence has been closing from both sides. On the theoretical side, models with nominal rigidities, transactions costs, and nontraded goods explain why real exchange rates can deviate substantially and persistently from PPP. On the empirical side, nonlinear econometric methods, longer data spans, and panel techniques provide stronger evidence of mean reversion, with plausible half-lives of 1–3 years for moderate shocks. The equilibrium real exchange rate itself may shift over time due to productivity differentials (Harrod-Balassa-Samuelson) and wealth effects, and allowing for such shifts further tightens estimates. The emerging consensus is that short-run PPP does not hold, but long-run PPP holds in a modified sense—real exchange rates revert toward a potentially time-varying equilibrium.

1.4 Key Takeaways

1. **PPP fails in the short run but holds approximately in the long run.** Neither absolute nor relative PPP holds continuously, but over decades, exchange rate changes track inflation differentials—especially when measured with producer price indices.
2. **The “PPP puzzles” are partly an artifact of linear methods.** Standard unit root tests lack power, and linear autoregressive models overestimate half-lives. Nonlinear models (ESTAR/TAR) that allow faster adjustment for larger deviations yield half-lives of 1–3 years rather than 3–5 years.
3. **Transactions costs create a band of inaction around PPP.** Goods arbitrage only corrects price differentials that exceed shipping and other frictions, producing nonlinear dynamics in which small deviations persist but large deviations revert more quickly.
4. **The equilibrium real exchange rate is not necessarily constant.** The Harrod-Balassa-Samuelson effect (productivity-driven trends in nontraded goods prices) and net foreign asset positions can cause the PPP-consistent real rate to drift over time. Failing to account for this biases reversion-speed estimates.
5. **Interdisciplinary convergence.** Transaction-cost insights now link international macroeconomics with international trade theory, and new-open-economy models with nominal rigidities help reconcile short-run volatility with long-run PPP.

2 Key Points

2.1 The PPP Concept: Absolute vs. Relative

Purchasing power parity exists in two forms. **Absolute PPP** asserts that the exchange rate E (domestic currency per unit of foreign currency) equals the ratio of price levels:

$$E = \frac{P}{P^*},$$

where P is the domestic and P^* the foreign aggregate price level. **Relative PPP** is the weaker claim that the *percentage change* in the exchange rate equals the inflation differential:

$$\frac{\Delta E}{E} \approx \pi - \pi^*,$$

where π and π^* are domestic and foreign inflation rates. Absolute PPP implies relative PPP, but the converse does not hold, since exchange rates could change proportionally even if the *levels* are misaligned (for example, due to persistent transactions costs).

The theoretical foundation for absolute PPP is the **Law of One Price** (LOP): identical traded goods should cost the same everywhere once expressed in a common currency, because arbitrageurs would otherwise ship goods from cheap to expensive locations. If the same goods enter each country's price index with the same weights, the LOP aggregates to absolute PPP. This link to [Key Point 2.9](#) on transactions costs is critical, since violations of the LOP are the most direct source of PPP failure.

2.2 Informal Empirical Evidence: Figures 1 and 2

The paper presents graphical evidence on the validity of PPP.

Figure 1 plots U.S. and UK consumer and producer price indices, both expressed in U.S. dollars, over roughly two centuries. Three lessons emerge: (i) absolute PPP does not hold perfectly at any point in time; (ii) the two price levels track each other over the very long run; and (iii) the co-movement is stronger for producer prices than for consumer prices.

Figure 2 uses data from a large cross-section of countries (1970–1998) to plot relative inflation against exchange rate depreciation. At the annual frequency, the scatter is wide—relative PPP clearly fails in the short run. However, when 29-year averages are taken, the scatter collapses onto the 45° line, confirming that relative PPP holds well over long horizons.

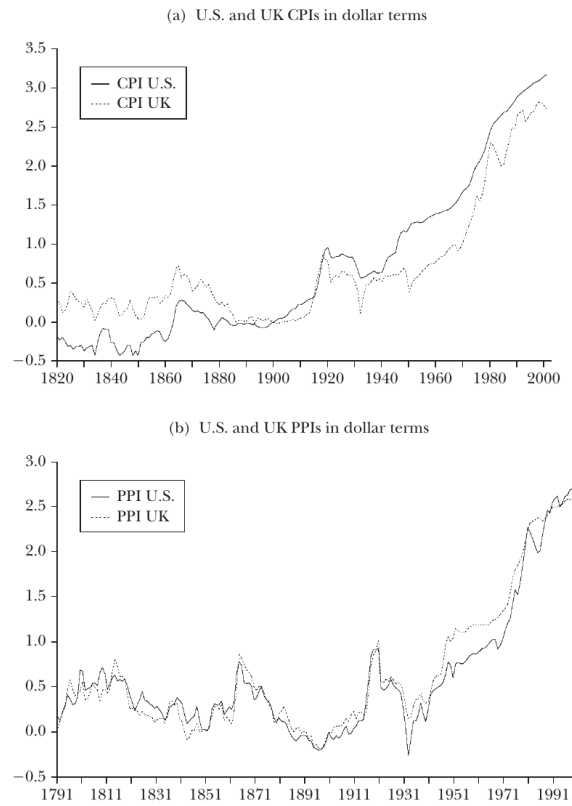
The overall lesson from [Key Point 2.1](#) and the graphical evidence is that PPP is best understood as a long-run anchor, not a short-run regularity, and that PPI-based measures provide tighter fits than CPI-based ones, consistent with the role of nontraded goods in CPIs.

2.3 The Monetary Approach and the Collapse of Continuous PPP

In the early floating-rate period after 1973, the dominant paradigm was the **monetary approach** to exchange rate determination, which assumed PPP held *continuously*. Under this view, the exchange rate—being the relative price of two monies—adjusts instantly as relative money supplies change, with the transmission running through relative goods prices and international arbitrage (linking back to the LOP mechanism in [Key Point 2.1](#)).

Early empirical tests of continuous PPP were encouraging, partly because the dollar was relatively stable in the first years of the float and the data span was short. By the late 1970s, as the dollar became far more volatile and longer data series became available, both continuous PPP and the simple monetary approach were decisively rejected. The key observation was that nominal exchange rates are far more volatile than relative national price levels, so real exchange rates move substantially in the short run—a pattern incompatible with continuous PPP. This episode, labeled the “collapse of purchasing power parity” by Frenkel (1981), set the stage for the subsequent focus on PPP as a *long-run* phenomenon (see [Key Point 2.4](#)).

Figure 1
Dollar-Sterling PPP Over Two Centuries



Notes: This figure shows U.S. and UK consumer and producer price indices expressed in U.S. dollar terms over roughly the last two centuries using a log scale with a base of 1900 = 0.

periods of ten years or so.) Thus, *relative* PPP seems to hold in a long-run sense. With the longer-run averages, the degree of correlation between relative inflation and exchange rate depreciation again seems higher when using PPIs than when using CPIs.

So the conclusions emerging from our informal eyeballing of Figures 1 and 2 seem to be the following. Neither absolute nor relative PPP appear to hold closely

Figure 1: **Figure 1 from the paper.** Dollar-Sterling PPP over two centuries. Panel (a) shows U.S. and UK CPIs in dollar terms; panel (b) shows PPIs. The closer co-movement of PPIs reflects the higher traded-goods content of the producer price index.

2.4 Random Walks, Unit Roots, and the Failure to Find Long-Run PPP

The theoretical response to the failure of short-run PPP was **exchange rate overshooting** (Dornbusch, 1976): prices are sticky, so monetary shocks cause the nominal exchange rate to jump beyond its long-run PPP level, with the real exchange rate reverting gradually as prices adjust. This retains PPP as the long-run equilibrium.

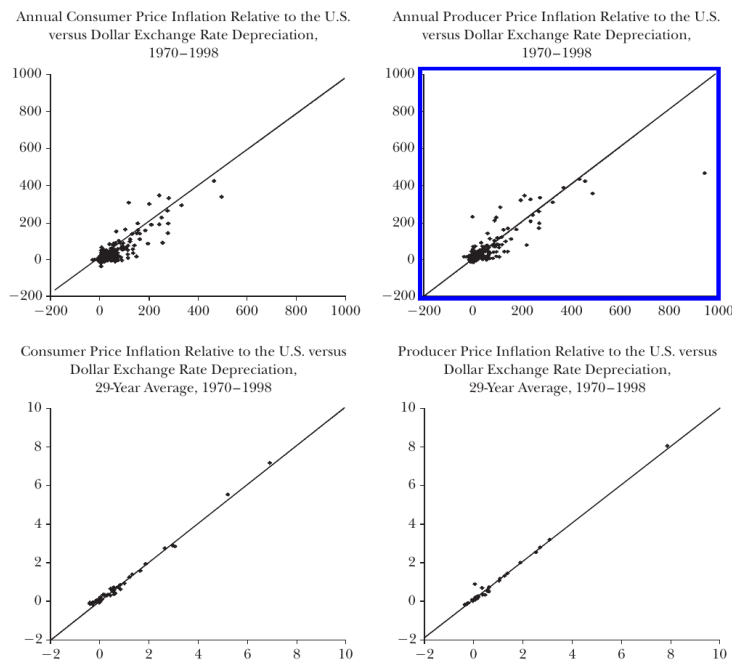
Testing long-run PPP empirically requires showing that the **real exchange rate** is mean-reverting. The real exchange rate is:

$$q_t = e_t + p_t - p_t^*,$$

where e_t is the log nominal exchange rate and p_t, p_t^* are log price levels. If PPP holds in the long run, q_t should revert to some constant (or slowly moving) mean.

Early tests (Roll, 1979; Adler and Lehmann, 1983) tested whether q_t follows a **random**

Figure 2

PPP at Various Time Horizons

Notes: This figure shows countries' cumulative inflation rate differentials against the United States in percent (vertical axis) plotted against their cumulative depreciation rates against the U.S. dollar in percent (horizontal axis). The charts on the left show CPI inflation, those on the right PPI inflation. The charts in the top row show annual rates, those in the bottom row 29-year average rates from 1970 to 1998.

In other words, as far as exchange rates are concerned, the fundamental things—relative price levels—apply increasingly as time goes by. These conclusions are in fact broadly in line with the current consensus view on PPP. So where does all the controversy arise?

The PPP Debate: A Tour of the Past Three Decades

Figure 2: **Figure 2 from the paper.** Relative inflation vs. exchange rate depreciation at annual (top row) and 29-year average (bottom row) frequencies, for CPI (left) and PPI (right). The convergence to the 45° line at long horizons demonstrates long-run relative PPP.

walk—the canonical non-mean-reverting process. The more formal unit root framework tests whether $\alpha = 1$ in:

$$q_t = \mu + \alpha q_{t-1} + \varepsilon_t.$$

If $\alpha = 1$, shocks to q_t accumulate permanently and q_t never settles at any particular level—long-run PPP fails even as a necessary condition.

Studies in the late 1980s (Taylor, 1988; Mark, 1990) uniformly *failed to reject* the unit root null for major real exchange rates. This was interpreted by many as evidence against long-run PPP, creating deep uncertainty. However, as [Key Point 2.5](#) explains, this result was most likely driven by the low statistical power of the tests.

2.5 The Power Problem

Frankel (1986, 1990) identified a critical flaw in the unit root literature: the tests employed only about 15 years of post-1973 data, and their statistical **power**—the probability of correctly rejecting a false null—was extremely low.

Simulations demonstrate the severity of this problem. Assuming mean reversion of roughly 11% per year (consistent with later estimates), the probability of rejecting a random-walk null at the 5% level is only about 5–8% with 15 years of data. Even with a century of data, the power is below 50%. Crucially, increasing the *frequency* of observation (e.g., moving from quarterly to monthly data) does not help, since higher-frequency data provides more information about short-run dynamics but not about long-run mean reversion (Shiller and Perron, 1985).

This insight motivated two strategies to gain statistical power:

- **More years:** long-span studies using a century or more of data ([Key Point 2.6](#));
- **More countries:** panel-data studies pooling information across exchange rates ([Key Point 2.7](#)).

2.6 Long-Span Studies

Researchers attacked the power problem by extending the time dimension. Using annual dollar-sterling data from 1869 to 1984, Frankel (1986) estimated:

$$q_t - \tilde{q} = \alpha (q_{t-1} - \tilde{q}) + \varepsilon_t,$$

where $\tilde{q}$ is the assumed equilibrium real exchange rate and α is the autocorrelation coefficient. Frankel obtained $\hat{\alpha} = 0.86$ and rejected the random walk at the 5% level. Similar results were found by Edison (1987, data 1890–1978), Glen (1992, data 1900–1987), and Lothian and Taylor (1996) using two centuries of dollar-sterling and franc-sterling data (with $\hat{\alpha} = 0.89$ and 0.76, respectively). Taylor (2002) extended the analysis to 20 countries over 1870–1996 and confirmed PPP with stable long-run coefficients.

A potential concern is that long spans encompass different exchange rate regimes (gold standard, Bretton Woods, floating), but Lothian and Taylor (1996) found no significant structural break between the pre- and post-Bretton Woods periods, lending robustness to the findings. These formal results provide the statistical counterpart to the graphical evidence in [Key Point 2.2](#).

2.7 Panel-Data Studies and Their Limitations

An alternative route to greater power is cross-sectional: pool many countries' real exchange rates for the post-Bretton Woods period. Abuaf and Jorion (1990) estimated a system of 10 autoregressive equations with a common α over 1973–1987 and marginally rejected joint non-mean-reversion. A large literature in the 1990s applied panel unit root tests and generally claimed support for long-run PPP.

However, Taylor and Sarno (1998) raised an important caveat. The standard panel unit root test has the null hypothesis that *none* of the real exchange rates are mean-reverting. Rejection therefore only implies that *at least one* rate is mean-reverting—yet researchers commonly inferred that *all* rates were, which is logically invalid. Testing the alternative null that at least one rate is non-mean-reverting (whose rejection would imply all are mean-reverting) requires less powerful tests, and results have been inconclusive.

This limitation means panel studies, while suggestive, do not conclusively establish universal long-run PPP for the recent float. Together with the long-span studies ([Key Point 2.6](#)), the evidence is interpreted as broadly supportive but far from airtight, motivating the identification of the PPP puzzles ([Key Point 2.8](#)).

2.8 The Two PPP Puzzles

The literature by the early 2000s had generated two distinct puzzles, as synthesized by Taylor, Peel and Sarno (2001) and Rogoff (1996):

Puzzle 1: Weak evidence for long-run PPP. Despite theoretical expectations, strong empirical evidence for long-run PPP—especially for the post-Bretton Woods period alone—has been elusive (see [Key Point 2.4](#)–[Key Point 2.7](#)).

Puzzle 2: Extremely slow adjustment speed. Even studies supporting mean reversion find that the **half-life** of real exchange rate shocks is 3–5 years. The half-life is computed from the autoregressive coefficient α : after n periods, a fraction α^n of the initial shock remains, and the half-life satisfies $\alpha^{n_{1/2}} = 0.5$. While real shocks (technology, tastes) could account for some persistent deviations, the bulk of real exchange rate movements appear driven by nominal shocks. Nominal wages and prices should adjust much faster than a 3–5 year half-life implies.

The key insight of the paper is that both puzzles may be artifacts of the **linear framework** conventionally employed, motivating the nonlinear approach discussed in [Key Point 2.9](#).

2.9 Nonlinear Adjustment and Transactions Costs

The paper’s central resolution of the PPP puzzles draws on the idea—dating to Heckscher (1916)—that transactions costs in international goods arbitrage induce **nonlinear dynamics**.

The intuition: If shipping goods between countries entails costs (transport, tariffs, time), arbitrage is only profitable when the price differential exceeds these costs. Small deviations from PPP persist because they fall within a “band of inaction,” while large deviations trigger arbitrage that pushes the real exchange rate back toward PPP. The adjustment speed therefore *increases with the size of the deviation*.

Threshold autoregressive (TAR) models formalize this by allowing α to switch between regimes depending on whether $|q_t - \tilde{q}|$ exceeds a threshold. For individual commodities, this works straightforwardly. Canjels, Prakash-Canjels and Taylor (forthcoming) applied a TAR model to dollar-sterling under the gold standard (1879–1913) and found results consistent with the framework. Obstfeld and Taylor (1997) modeled price adjustment across international cities post-1973 and found significant nonlinearities with transaction-cost bands of plausible size.

For *aggregate* real exchange rates, however, there is a conceptual difficulty: different goods have different transactions costs, so there is no single threshold. As the real exchange rate deviates further from PPP, more goods breach their individual thresholds, so arbitrage pressure increases smoothly. This motivates **smooth transition autoregressive (STAR) models**—specifically the **exponential STAR (ESTAR)** model—which allow the autoregressive coefficient to vary continuously as a function of the deviation size.

Taylor, Peel and Sarno (2001) applied the ESTAR model to G-5 real dollar exchange rates for the post-Bretton Woods period and achieved two key results:

- They rejected the unit root null in favor of nonlinear mean reversion *using only post-1973 data*—solving **Puzzle 1**.
- For modest shocks (1–5%), the estimated half-life was under 3 years; for larger shocks, it was considerably shorter—going a substantial way toward solving **Puzzle 2**.

Moreover, simulations show that if the true process is nonlinear, fitting a linear unit root test severely reduces power, and estimated half-lives are dramatically exaggerated. This means the disappointing results of the linear literature ([Key Point 2.4](#)) are exactly what one would expect if the true process were nonlinear all along.

Other sources of nonlinearity have also been proposed: heterogeneity of expectations among foreign exchange traders (Kilian and Taylor, 2003) and central bank intervention, which is more likely and effective at extreme exchange rate values (Taylor, 2004).

2.10 Short-Run Exchange Rate Volatility

Even if nonlinear models improve medium-run reversion speed estimates, short-run exchange rate volatility remains a puzzle. Over short horizons (months), nominal exchange rates move substantially while prices barely move, so nominal and real exchange rate volatility are almost perfectly correlated. This pattern persists across monetary regimes.

Several factors contribute. First, economies may be more “closed” than traditionally assumed: large sectors like retailing, wholesaling, and distribution are nontraded, so the effective nontraded share of the economy exceeds 50%. In models with price stickiness and a large nontraded sector, even small monetary shocks generate large exchange rate movements (Burstein, Eichenbaum and Rebelo, 2003). Second, a larger nontraded share implies a smaller fraction of goods subject to the LOP, amplifying the scope for exchange rate overshooting (Obstfeld and Rogoff, 2000; Hau, 2000, 2002). These insights link to the nontraded-goods channel central to [Key Point 2.11](#).

2.11 The Harrod-Balassa-Samuelson Effect and Trending Equilibrium

The standard PPP framework assumes the equilibrium real exchange rate is *constant*. The Harrod-Balassa-Samuelson (HBS) model explains why it may *trend* over time, providing a crucial modification.

The mechanism: Rich countries grow rich through productivity gains in traded (“modern”) sectors such as manufacturing. If the Law of One Price holds for tradables and labor is mobile across sectors (but not across countries), wages rise economy-wide. Since the nontraded (“traditional”) sector—haircuts, services—has experienced no productivity gain, its prices must rise to cover higher wages. The aggregate price level (a weighted average of traded and nontraded prices) therefore rises relative to that of poorer countries, making richer countries appear “overvalued” in PPP terms. The prediction: a positive cross-country correlation between per capita income and the price level.

As Figure 3 shows, this correlation has strengthened markedly since 1950. The elasticity was close to zero and statistically insignificant in 1913, but exceeded 0.5 by 1995. Why the effect has intensified over time is debated; Bergin, Glick and Taylor (2004) suggest that trade costs endogenously determine which goods are traded, so productivity shocks can generate an HBS effect whose magnitude evolves.

Implications for PPP estimation: If the equilibrium real rate trends and we estimate a model assuming it is fixed, the estimated reversion speed is biased downward (the half-life is inflated). Taylor (2002) finds lower half-lives in a 20-country panel when allowing for long-run trends. Lothian and Taylor (2000) report a half-life as low as $2\frac{1}{2}$ years for dollar-sterling when allowing for nonlinear time trends.

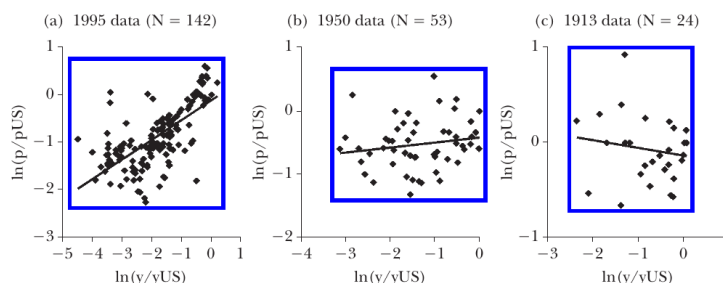
2.12 Net Foreign Asset Positions and the Real Exchange Rate

The second channel through which the equilibrium real exchange rate may shift is the **net international asset position**. If a country accumulates external debt, it must run future trade surpluses to service interest payments. Generating a surplus requires improved competitiveness—a real depreciation (rise in q when defined as domestic price of foreign currency times the ratio P/P^*). Hence, larger net external debt implies a weaker equilibrium real exchange rate.

Lane and Milesi-Ferretti (2002) provide empirical support: countries with larger positive net foreign asset positions have stronger real exchange rates (and more negative trade balances), controlling for other factors. If these wealth-driven shifts in the equilibrium rate are not accounted for, the real exchange rate appears to deviate from a “fixed” PPP rate for longer than it actually does, further contributing to Puzzle 2 ([Key Point 2.8](#)).

Figure 3

Harrod-Balassa-Samuelson Effects Emerge: Log Price Level versus Log Per Capita Income



Notes: This figure shows countries' log price level (vertical axis) against log real income per capita for 1995, 1950 and 1913, with the United States used as the base country.

Samuelson effect has been variable over time—certainly in the postwar period and perhaps going back several centuries (Bergin, Glick and Taylor, 2004). Consider the relationships in Figure 3. The horizontal axis shows the per capita income level of countries as a ratio of the U.S. per capita income level, expressed in log terms. The vertical axis shows the common currency Penn World Table Consumer Price Index price level of other countries as a ratio of the U.S. CPI price level—that is, the real exchange rate level—again expressed as a log. (Note that most countries have incomes and price levels lower than the United States, so the ratios are less than one, and the logs are negative.) The Harrod-Balassa-Samuelson hypothesis suggests

Figure 3: **Figure 3 from the paper.** Log price level vs. log per capita income (relative to the U.S.) for 1995, 1950, and 1913. The positive slope—the HBS effect—has been intensifying over time, with the elasticity now exceeding one half.

2.13 Engel's Decomposition and Unifying the Puzzles

Engel (1999, 2000) decomposed real exchange rate variance into two components: (i) the deviation of traded goods prices across countries (failures of the LOP), and (ii) the relative price of traded vs. nontraded goods within countries (the HBS channel). He found that both components are persistent, but the LOP component experiences larger shocks.

This decomposition connects the various strands of the paper. The LOP component relates to the standard PPP theory where the real rate is assumed fixed, and is best studied using nonlinear models with trade costs (Key Point 2.9). The nontraded-goods component relates to the HBS effect (Key Point 2.11) and models the slower drift. Recent models of endogenous tradability (Betts and Kehoe, 2001; Bergin and Glick, 2003) attempt to unify both channels.

3 Logical Chains

3.1 Chain 1: From Transactions Costs to Resolution of the PPP Puzzles

Step 1. International goods arbitrage is costly: Transport costs, tariffs, time lags, and nontariff barriers mean that shipping goods between countries incurs nontrivial expense.

Explanation: This is an empirical premise, well documented in the trade literature (Anderson and van Wincoop, 2004). These costs drive a wedge between domestic and foreign goods prices.

Step 2. Small price differentials are not arbitrated: If the expected profit from shipping goods does not exceed the transactions cost, no trade occurs. This creates a “band of inaction” around PPP within which real exchange rate deviations persist.

Explanation: Follows directly from Step 1 by a simple profitability calculation: arbitrage requires $|P - E \cdot P^*| > c$, where c represents costs.

Step 3. Larger deviations trigger arbitrage and faster correction: As the real exchange rate moves further from PPP, more goods breach their individual cost thresholds, increasing the volume and speed of corrective trade.

Explanation: Different goods have different transactions costs. As the aggregate deviation grows, the fraction of goods whose thresholds are exceeded increases monotonically, generating smooth nonlinear mean reversion.

Step 4. The aggregate adjustment speed is a nonlinear function of deviation size: In an ESTAR or TAR model, the autoregressive coefficient α varies with $|q_t - \tilde{q}|$, being close to 1 for small deviations (slow/no reversion) and well below 1 for large deviations (fast reversion).

Explanation: This is the econometric formalization of Step 3, aggregating heterogeneous thresholds into a smooth transition.

Step 5. Linear tests are misspecified and low-powered against this alternative: If the true process is ESTAR, a linear unit root test loses power because it effectively averages over fast and slow adjustment regimes, biasing $\hat{\alpha}$ toward 1 and inflating the half-life.

Explanation: Simulations by Taylor, Peel and Sarno (2001) confirm this: estimated half-lives under linear misspecification are dramatically exaggerated.

Conclusion. Both PPP puzzles are substantially resolved. (i) The failure to reject unit roots (Puzzle 1) reflects low power of linear tests against a nonlinear alternative, not the absence of mean reversion. (ii) The slow estimated adjustment speed (Puzzle 2) is an artifact of imposing a single half-life on a process where large shocks revert in 1–3 years and small shocks revert very slowly.

Explanation: Steps 1–5 combine to show that the puzzles arise from applying the wrong econometric framework to a process whose true dynamics are governed by heterogeneous arbitrage costs.

3.2 Chain 2: The Harrod-Balassa-Samuelson Mechanism

Step 1. Productivity growth is concentrated in the traded sector: Rich countries grow rich primarily via advances in manufacturing and other tradable industries; the non-traded sector (services) sees little productivity growth.

Explanation: This asymmetry is an empirical regularity; technological progress is more rapid in goods production than in services.

Step 2. The Law of One Price holds for traded goods: International arbitrage keeps traded-goods prices roughly equalized across countries (in common currency).

Explanation: Standard arbitrage argument from [Key Point 2.1](#), subject to the transactions-cost caveats of [Key Point 2.9](#).

Step 3. Higher traded-sector productivity raises economy-wide wages: If labor is mobile between sectors within a country, wages in the nontraded sector must match those in the traded sector.

Explanation: In a competitive labor market, intersectoral wage equalization occurs through worker mobility.

Step 4. Nontraded goods prices rise: With unchanged productivity but higher wages, the nontraded sector must charge higher prices to cover costs.

Explanation: Cost-push: unit labor costs in the nontraded sector rise one-for-one with wages since there has been no offsetting productivity gain.

Step 5. The aggregate price level rises relative to less productive countries: Since the overall CPI is a weighted average of traded and nontraded goods prices, and traded-goods prices are equalized, the richer country's CPI is higher.

Explanation: The nontraded component drives up the aggregate price level, creating an apparent "overvaluation" relative to PPP.

Conclusion. The equilibrium real exchange rate appreciates with productivity growth. This trend means that the "PPP rate" is a moving target, and any regression framework that assumes a fixed equilibrium will underestimate the speed of mean reversion and overestimate the half-life.

Explanation: Steps 1–5 build the HBS mechanism; the econometric implication is that omitting a time-varying equilibrium biases estimates, contributing to Puzzle 2 ([Key Point 2.8](#)).

4 Key Terms & Definitions

Absolute Purchasing Power Parity (Absolute PPP)

The proposition that the nominal exchange rate equals the ratio of domestic to foreign aggregate price levels, $E = P/P^*$, so that a unit of currency has identical purchasing power in both countries.

Band of Inaction

A range around the PPP-consistent real exchange rate within which deviations persist because they are too small to justify the transactions costs of international goods arbitrage. Within this band, price dynamics in different locations are essentially disconnected.

Bretton Woods System

The post-World War II international monetary arrangement (1944–1971) in which the U.S. dollar was pegged to gold and other currencies were pegged to the dollar. Its collapse in 1971–73 ushered in the era of floating exchange rates.

ESTAR (Exponential Smooth Transition Autoregressive) Model

A nonlinear time series model in which the speed of mean reversion increases smoothly with the absolute size of the deviation from equilibrium. Applied to real exchange rates, it captures the idea that larger PPP deviations trigger more arbitrage and faster correction. Formally, the transition function is exponential in $(q_{t-1} - \tilde{q})^2$.

Half-Life of Shocks

The number of periods n required for the effect of a shock to the real exchange rate to decay by 50%. In a linear AR(1) model with coefficient α , $n_{1/2} = \ln(0.5)/\ln(\alpha)$. The consensus estimate from linear models is 3–5 years; nonlinear models yield 1–3 years for moderate shocks.

Harrod-Balassa-Samuelson (HBS) Effect

The prediction that countries with higher productivity in traded goods will have higher relative prices of nontraded goods and thus higher overall price levels, causing their currencies to appear “overvalued” relative to simple PPP. This implies a trending equilibrium real exchange rate.

Law of One Price (LOP)

The principle that an identical traded good should sell for the same price everywhere when expressed in a common currency, enforced by arbitrage. The LOP is the microeconomic foundation of PPP, though it is violated in practice due to transactions costs, product differentiation, and nontradability.

Monetary Approach to the Exchange Rate

The 1970s framework in which the exchange rate, as the relative price of two monies, is determined by relative money market equilibria and PPP is assumed to hold continuously.

Power (of a statistical test)

The probability that a test correctly rejects a false null hypothesis. In the PPP context, the power of unit root tests applied to real exchange rates over short spans (15 years) is very low (5–8%), meaning failure to reject a random walk is uninformative about whether long-run PPP actually holds.

Real Exchange Rate

Defined as $q = e + p - p^*$ (in logs) or $Q = E \cdot P/P^*$ (in levels), where E is the nominal

exchange rate. It measures relative purchasing power: if PPP holds perfectly, $Q = 1$ (or q is constant). Variation in q indicates PPP deviations.

Relative Purchasing Power Parity (Relative PPP)

The weaker proposition that the rate of change of the nominal exchange rate equals the inflation differential between two countries. Does not require that *levels* of purchasing power be equalized, only that *changes* match.

Threshold Autoregressive (TAR) Model

A piecewise-linear time series model with regime switching: the autoregressive coefficient changes depending on whether the real exchange rate lies inside or outside a threshold (“commodity points”). Outside the band, arbitrage drives reversion; inside, movements are random.

Unit Root Process

A time series process in which $\alpha = 1$ in $q_t = \mu + \alpha q_{t-1} + \varepsilon_t$. Shocks accumulate permanently, and the series never settles at any level. A random walk (with or without drift) is the canonical example. Finding a unit root in the real exchange rate implies that long-run PPP does not hold.